MoonWalker Weekly Meeting

10/22 - 10/29 Summary

Elijah, Rowel, Martin

Progress Update

- Rowel
  - Connected MySQL Database to Web-Form-App
    - Completed Registration Test in User Manual
  - Fixed creating Credential in the Web-Form-App
- Martin & Elijah
  - Martin: Fix Unity Lunar terrain issue
  - Connected Unity and Omniverse
    - Viper and Cadre able to Produce & Consume live using HSML API, MySQL, and Kafka
  -

Generated HSML JSON


```
{
  "@context": "https://digital-twin-interoperability.github.io/hsml-schema-context/hsml.jsonld",
  "@type": "Person",
  "name": "Rowel Espejo",
  "birthDate": "2003-12-31",
  "email": "owelbespejo@gmail.com",
  "swid": "did:key:6MksTfieVPyUdcKEwRWggbizj4SoLHaQTnpAjFvjuLuvr7F"
}
```


Download JSON | Download mykey.pem

Back to Home

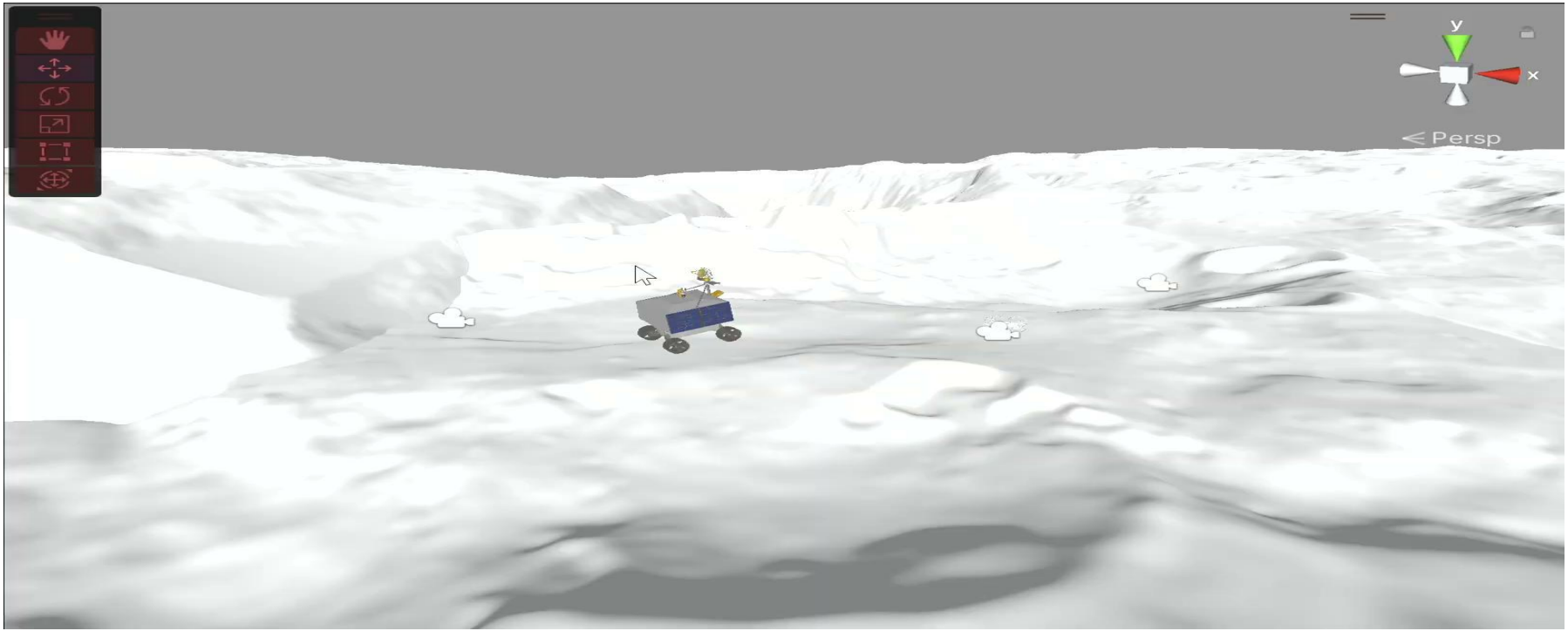
Kafka | Docker Contair | MySQL | HSML-Form-W | Ubuntu

```
mysql> SELECT * FROM did_keys ORDER BY did DESC LIMIT 1;
+----+-----------------------------------------------------------------+-------------------------------------------------------------------
-------------+-----------------------------------------------------------+-------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
--------+--------------------+------------+------------+
| id | did                                                             | registered_by
| public_key                                                | metadata


        | created_at          | kafka_topic | allowed_did |
+----+-----------------------------------------------------------------+-------------------------------------------------------------------
-------------+-----------------------------------------------------------+-------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
--------+--------------------+------------+------------+
|  1 | did:key:6MksTfieVPyUdcKEwRWggbizj4SoLHaQTnpAjFvjuLuvr7F | did:key:6MksTfieVPyUdcKEwRWggbizj4SoLHaQTnp
AjFvjuLuvr7F | 6MksTfieVPyUdcKEwRWggbizj4SoLHaQTnpAjFvjuLuvr7F | {"name": "Rowel Espejo", "swid": "did:key:6
MksTfieVPyUdcKEwRWggbizj4SoLHaQTnpAjFvjuLuvr7F", "@type": "Person", "email": "owelbespejo@gmail.com", "@cont
ext": "https://digital-twin-interoperability.github.io/hsml-schema-context/hsml.jsonld", "birthDate": "2003-
12-31"} | 2025-10-23 13:38:14 | NULL        | NULL        |
+----+-----------------------------------------------------------------+-------------------------------------------------------------------
-------------+-----------------------------------------------------------+-------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------------------
--------+--------------------+------------+------------+
1 row in set (0.00 sec)

mysql>
```


```
  }
],
"position": [
  {
    "@type": "schema:PropertyValue",
    "name": "xCoordinate",
    "value": 11.16111073821449
  },
  {
    "@type": "schema:PropertyValue",
    "name": "yCoordinate",
    "value": 13.40089505536677
  },
  {
    "@type": "schema:PropertyValue",
    "name": "zCoordinate",
    "value": -4.002000392258992
  }
],
"rotation": [
  {
    "@type": "schema:PropertyValue",
    "name": "rx",
    "value": 0.031422875035527945
  },
  {
    "@type": "schema:PropertyValue"
```

Fixing Lunar Floor Interaction - Last Week

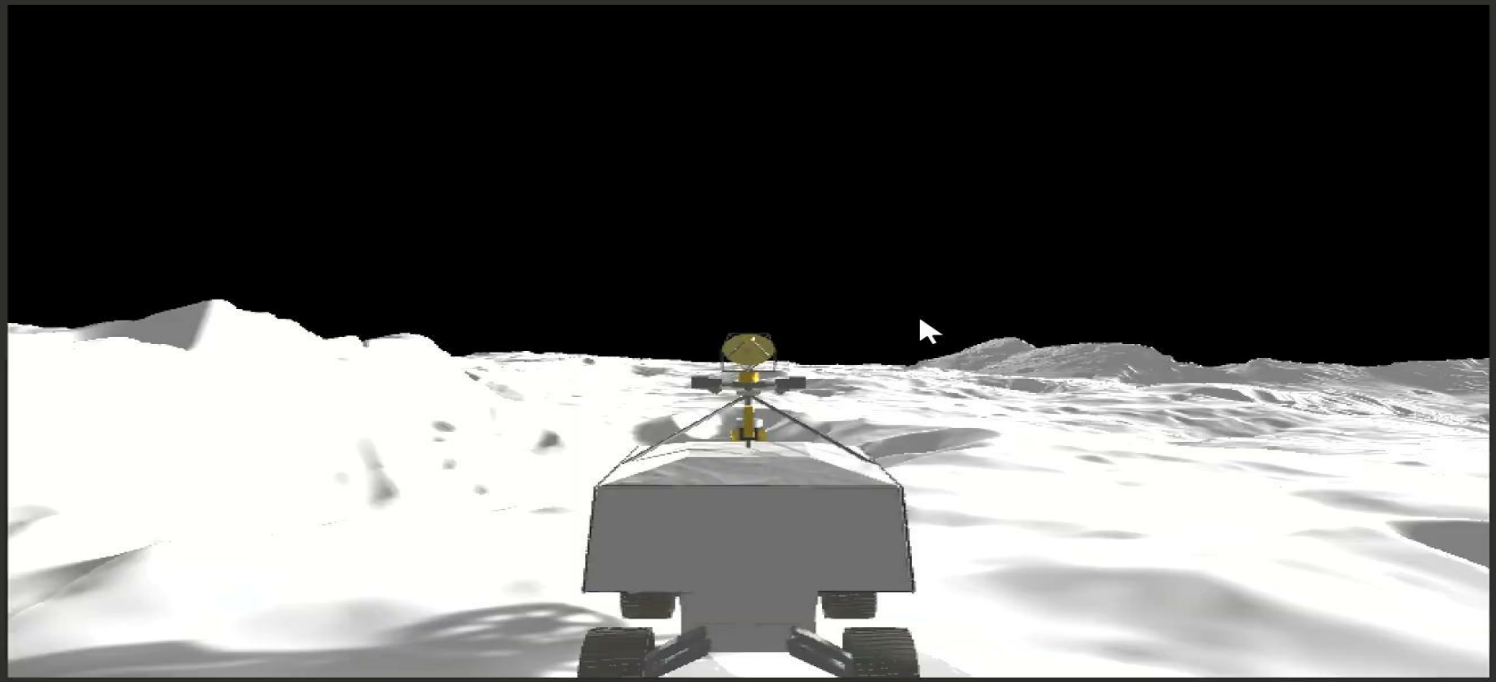
Lunar Terrain Problem: Enabling Gravity



Lunar Terrain Problem

- There isn’t a Shackleton ridge for Unity ready for Unity in the drive. Only Omniverse.
- Process:
  - Download Shackleton ridge .USD file
  - Download Blender
  - Convert .USF file to .FBX
- Import back into Unity
  - This creates issues with terrain settings and how Unity process such large assets with Mesh Collider
  - Specifically settings: Cook for Faster Simulation, **Enable Mesh Cleaning**, **Weld Colocated Vertices**, & Use Fast Midphase

Fix: Enable Mesh Cleaning and Weld Colocated Vertices



Next: Test 2 - HSML API Integration

- Issue: HSML API Credential Not working

```
  },
  "registered_by": "did:key:6Mkn3yTDH6Pba4upbZPUmPijuJETcKYEMPrVhKWDxQrGm5i",
  "domain_pem": "-----BEGIN PRIVATE KEY-----\nMC4CAQAwBQYDK2VwBCIEIFZWVyn1d3A2eZOR84ebManXptD7iwmpluxp56dU5x
ri\n-----END PRIVATE KEY-----\n"
}
DEBUG: API Response Status: 403
DEBUG: API Response Text: {"detail":"The 'issuedBy.swid' must match the DID of the user registering this Cre
dential."}
DEBUG: API call failed with status 403
172.26.16.1 - - [29/Oct/2025 07:56:00] "POST /create/credential HTTP/1.1" 302 -
DEBUG CREDENTIAL: Method=GET, DID in session=did:key:6Mkn3yTDH6Pba4upbZPUmPijuJETcKYEMPrVhKWDxQrGm5i, Form k
eys=N/A
172.26.16.1 - - [29/Oct/2025 07:56:00] "GET /create/credential HTTP/1.1" 200 -
172.26.16.1 - - [29/Oct/2025 07:56:00] "GET /static/css/style.css HTTP/1.1" 304 -
172.26.16.1 - - [29/Oct/2025 07:56:00] "GET /static/css/images/background.png HTTP/1.1" 304 -
```

Solution: Test 2 - HSML API Integration

Generated HSML JSON

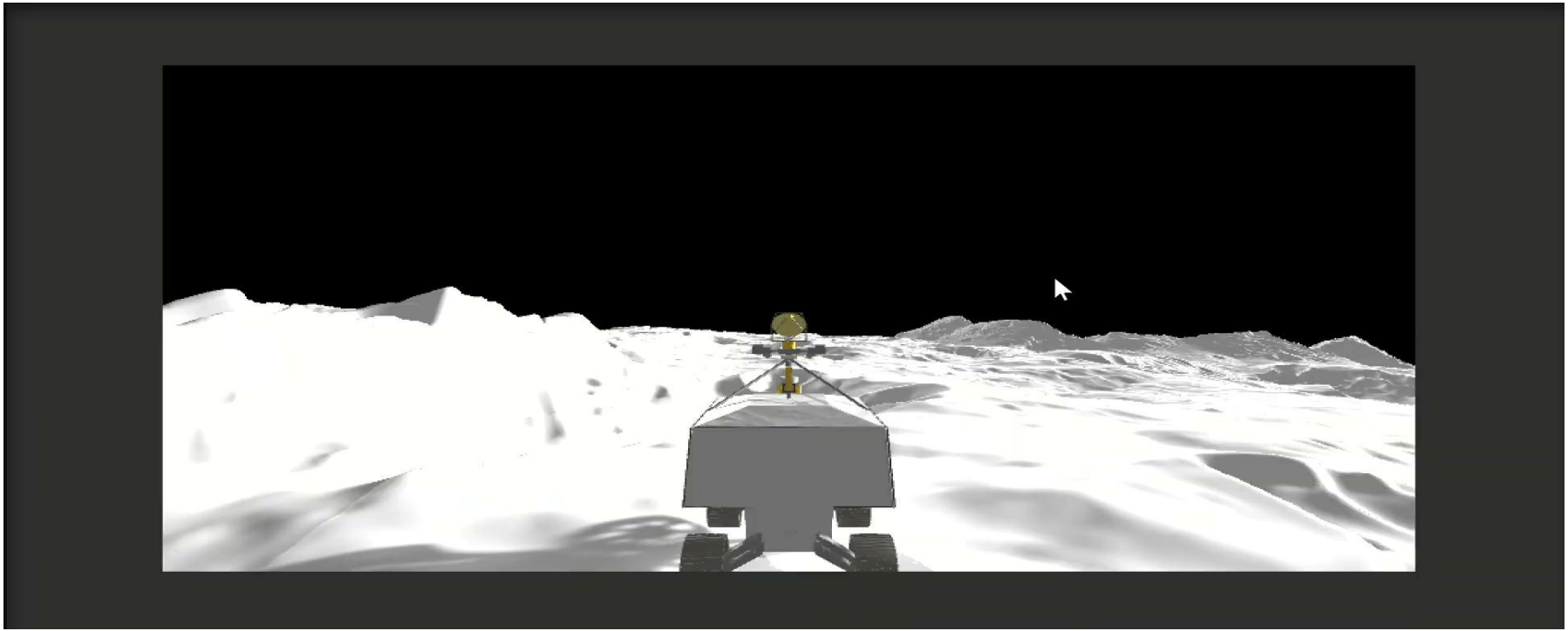

```
{
  "@context": "https://digital-twin-interoperability.github.io/hsml-schema-context/hsml.jsonld",
  "@type": "Credential",
  "name": "Example Agent 2 Access Credential ",
  "description": "Example Credential",
  "issuedBy": {
    "@type": "Person",
    "name": "John Smith",
    "swid": "did:key:6MkjKLczYJYYzj9oVuhetxVrSir7UjtQxru8tRWKiAPzH9D"
  },
  "accessAuthorization": {
    "@type": "Agent",
    "name": "Agent2",
    "swid": "did:key:6MkvoPsvcq12tHrGWreNW3dajKNF96DGa2gqNB142zJWqXu"
  },
  "authorizedForDomain": {
    "@type": "Agent",
    "name": "Agent1",
    "swid": "did:key:6MkevNxKVvCmrZ3G5NPN7bsDfukHV5Sg92ueV2Bha4vs93f"
  },
  "swid": "did:key:6MksXDp55oKpNDfNqmELwNWF1sNn3fPjWeCuuiKoKehZvgb"
}
```


Create Credential HSML Object

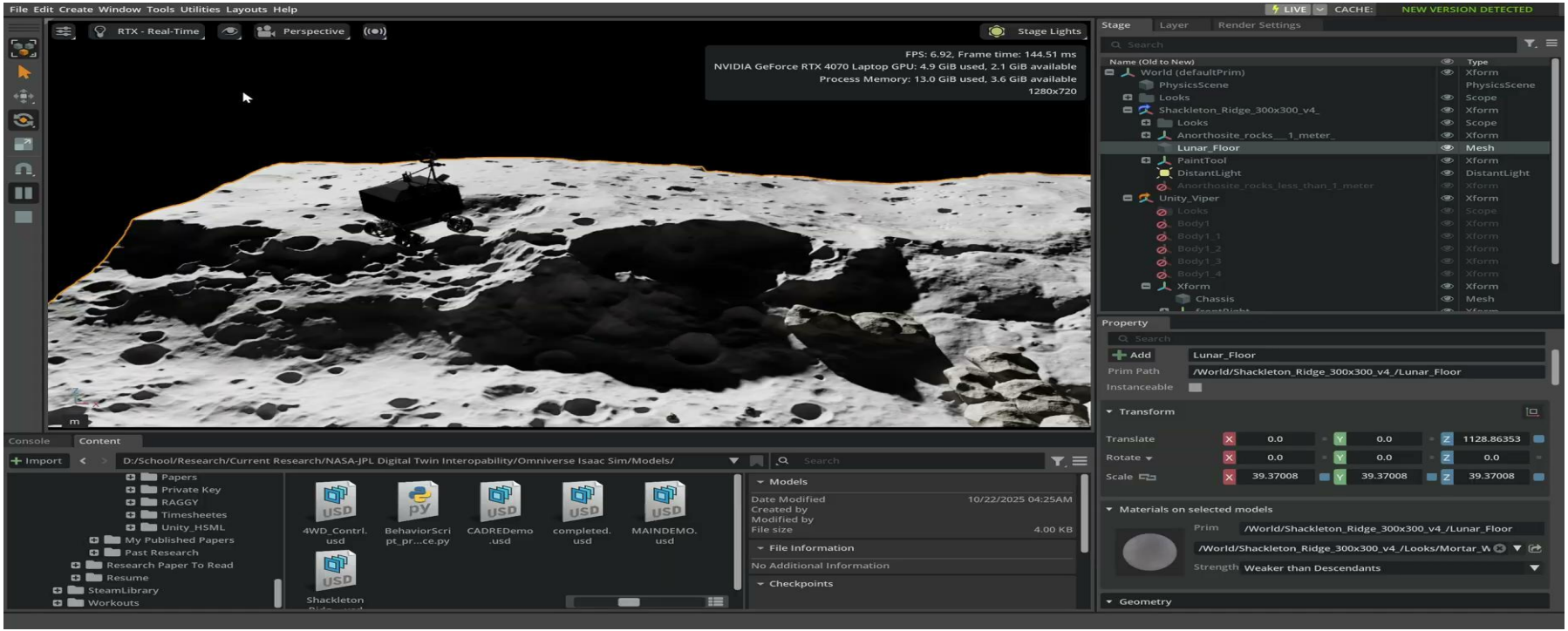
Docker Contair | Web-App | MySQL | Ubuntu | api logs


```
DEBUG: Payload: {
  "entity": {
    "@context": "https://digital-twin-interoperability.github.io/hsml-schema-context/hsml.jsonld",
    "@type": "Credential",
    "name": "omni_unity_cadrea_credential",
    "description": "test",
    "issuedBy": {
      "@type": "Person",
      "name": "Elijah",
      "swid": "did:key:6Mkpm86HYpbVbctEbgef74hby34ck9BDgQkCSm4hPt2FbAv"
    },
    "accessAuthorization": {
      "@type": "Agent",
      "name": "Elijah Omni",
      "swid": "did:key:6MksLzYr9CEkwZeAsaXe1nNEUeJ56pAiC6QwEZBjgC8SuZT"
    },
    "authorizedForDomain": {
      "@type": "Agent",
      "name": "Martin Unity",
      "swid": "did:key:6MktZdWk5BE6FEZMxtcS2iVD8QCeZ7N4P62mrPngm1E5shP"
    }
  },
  "registered_by": "did:key:6Mkpm86HYpbVbctEbgef74hby34ck9BDgQkCSm4hPt2FbAv",
  "domain_pem": "-----BEGIN PRIVATE KEY-----\nMC4CAQAwBQYDK2VwBCIEIFZWVyn1d3A2eZOR84ebManXptD7iwmpluxp56dU5x
ri\n-----END PRIVATE KEY-----\n"
}
DEBUG: API Response Status: 200
DEBUG: API Response Text: {"status":"success","did":"did:key:6MkhL7qZDRETBkHQ3gPYAsHptbGEdGnQkUosJzkY7GL15ug
","entity_type":"Credential","metadata":{"@context":"https://digital-twin-interoperability.github.io/hsml-sc
```

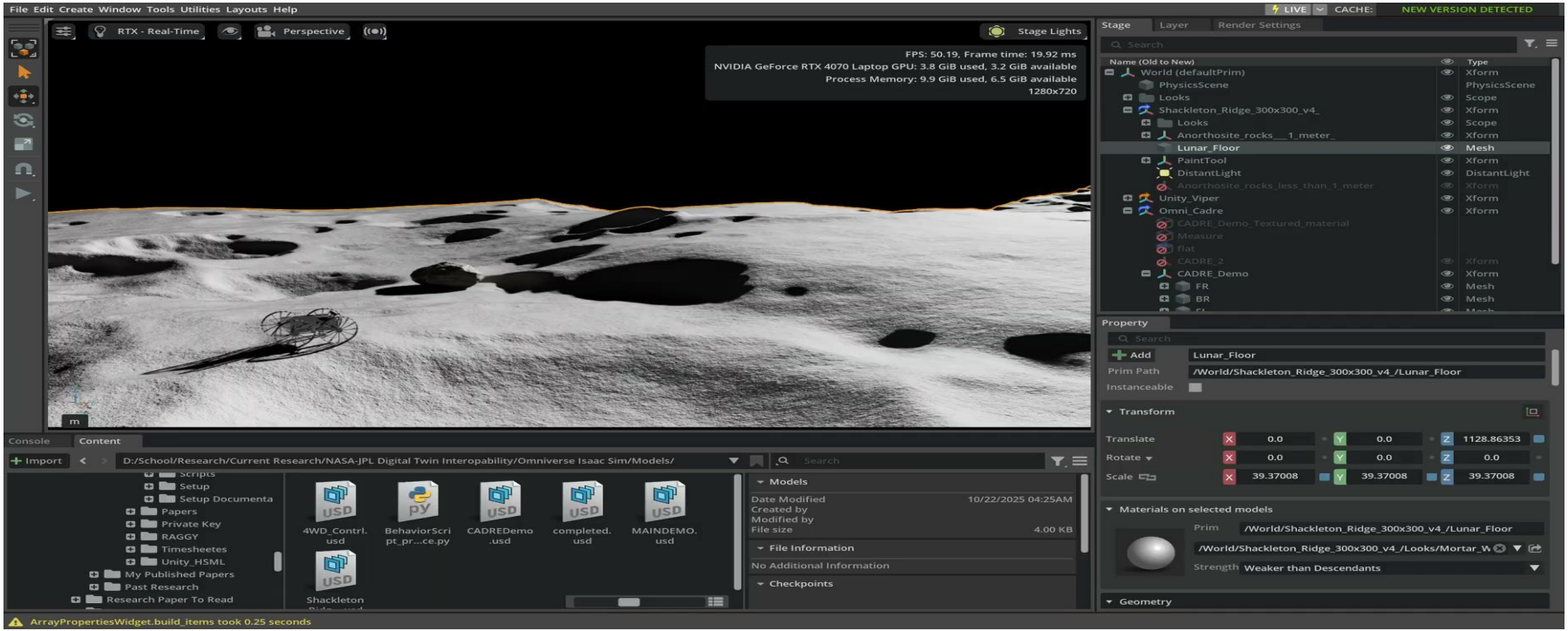
Initial Test 2: Unity - Viper (Host)



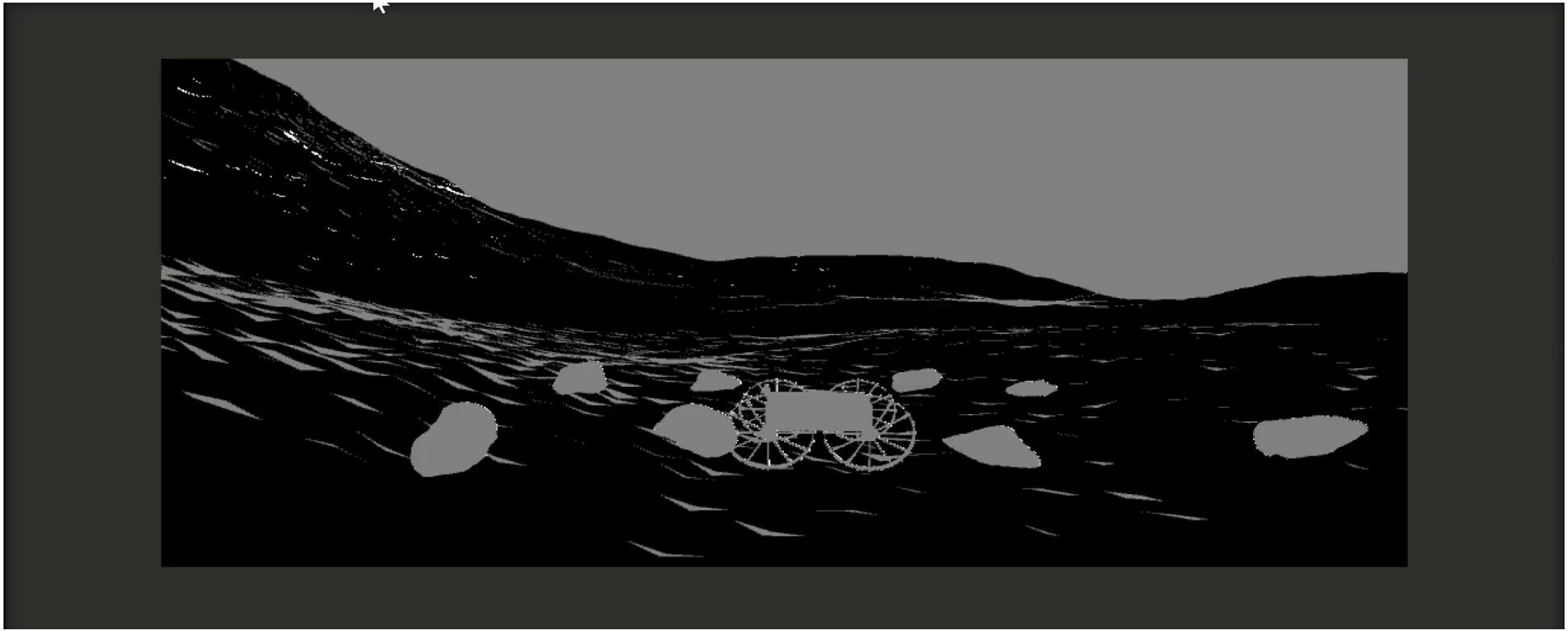
Initial Test 2: Omniverse - Viper (Chost)



Initial Test 2: Omni - Cadre (Host)

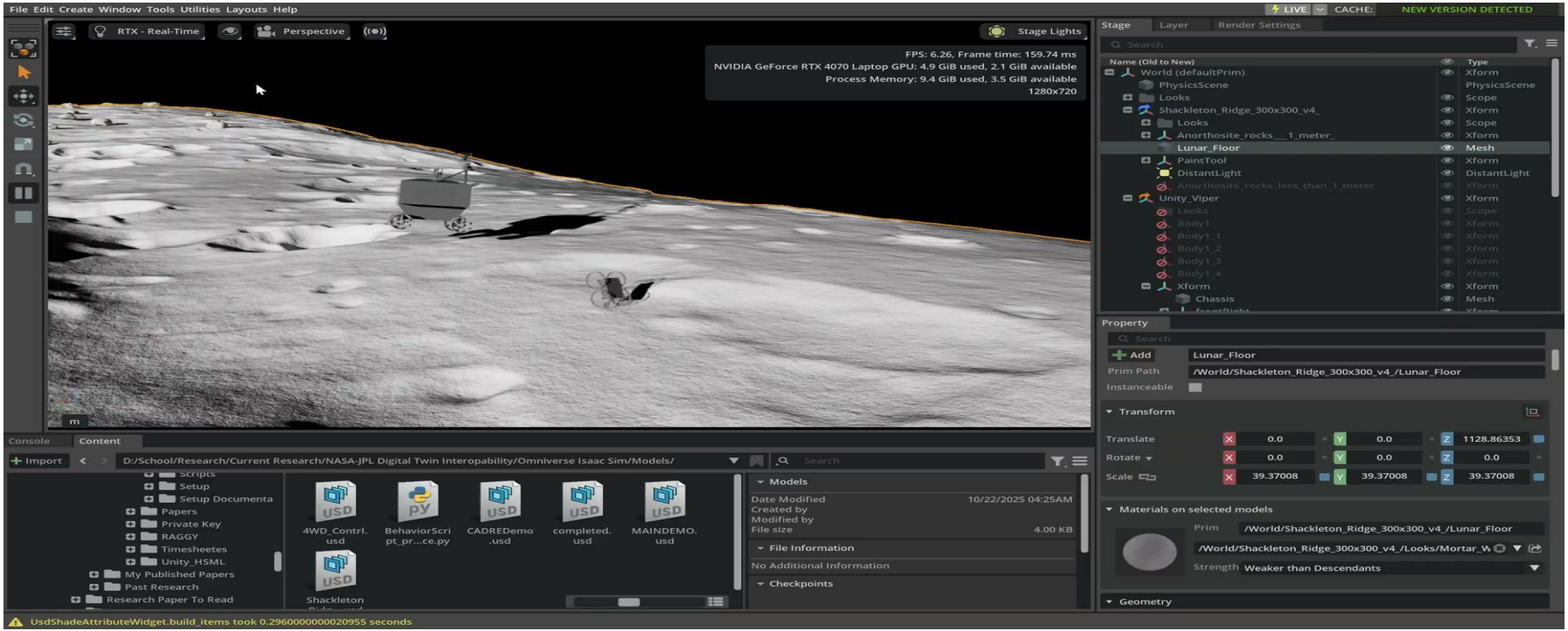


Initial Test 2: Unity - Cadre (Ghost)



Problems:

- The rovers spin around like crazy when simulation starts
  - When we press “play” on our simulations, the rovers spin around and teleport like crazy.
  - Hypothesis: We believe this has to due with how the API consumer scripts work when one simulation is turned on or at the same time as the other.
  - Not a significant problem as waiting around ~10 seconds, the rovers stop acting up.

- The rover coordinates aren’t precise between both simulations.
  - Comparing the HSML API scripts w/ Unity-Omniverse-Kafka scripts: the way coordinates are transformed between the two are a bit different
  - Identified discrepancies:
    - Floating Rovers; We believe this is due to how we both import ShackletonRidge
    - Sending data from Unity to Omni requires little transformation(Switch Y/Z)
    - Sending Data from Omni to Unity require more transformation

Further Testing: Test 2 (HSML API)